

3D ROBOTIC ELBOW REHABILITATION STIMULATOR

INTRODUCTION: This invention relates to the field of medical devices and physiotherapy tools for orthopaedic rehabilitation. This orthopedic monitoring system is designed for patients recovering from elbow surgery or trauma. Utilizing advanced computer vision, the software analyzes joint mobility and healing indicators to provide real-time feedback. It serves as a digital bridge between patients surgical recovery and physical rehabilitation, ensuring patients progress stays on track & allows safe home-based exercises under remote supervision.

Technical Field: This invention relates to an **Interactive Musculoskeletal Rehabilitation System**. It provides a 3D simulation environment for the guided physiological recovery of the elbow joint, specifically focused on **flexion and extension** protocols.

Detailed Description of Invention : The "3D Robotic Elbow Rehabilitation Stimulator" is an **Integrated Functional Interface** designed for high-precision elbow recovery. Utilizing a **Dynamic Kinematic Model**, the system allows patients to safely visualize and execute controlled movements to restore joint mobility after surgery or trauma. The patient sits on a chair or bed with the affected arm supported on a table or lap. The patient accesses the app through any standard web browser on a mobile phone or laptop. The robotic arm assists elbow flexion/extension movements (0° – 150°). The system provides real-time guidance to improve mobility and reduce errors.

Clinical challenge: Elbow flexion/extension recovery after trauma/surgery

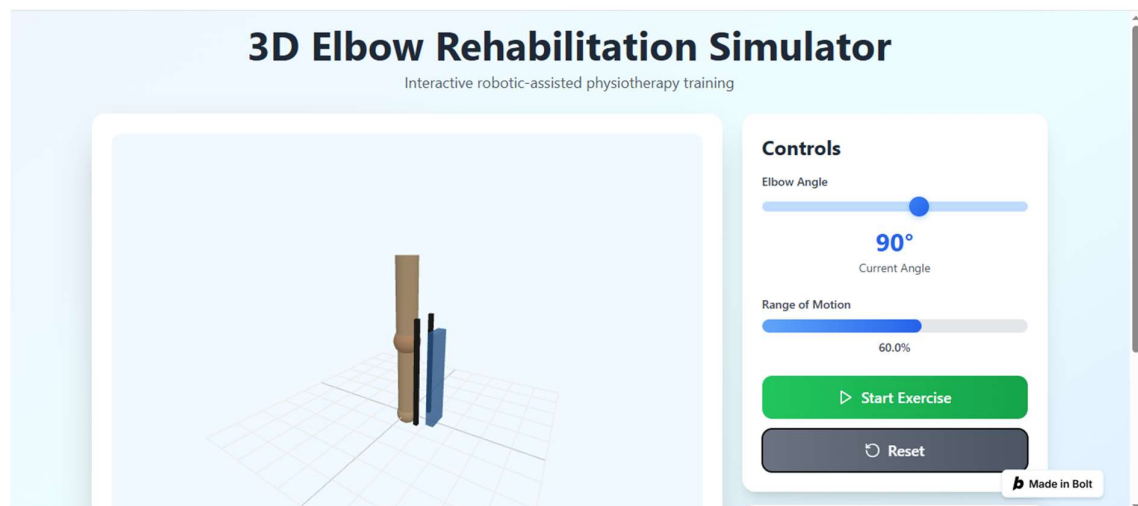
Key Functional Components:

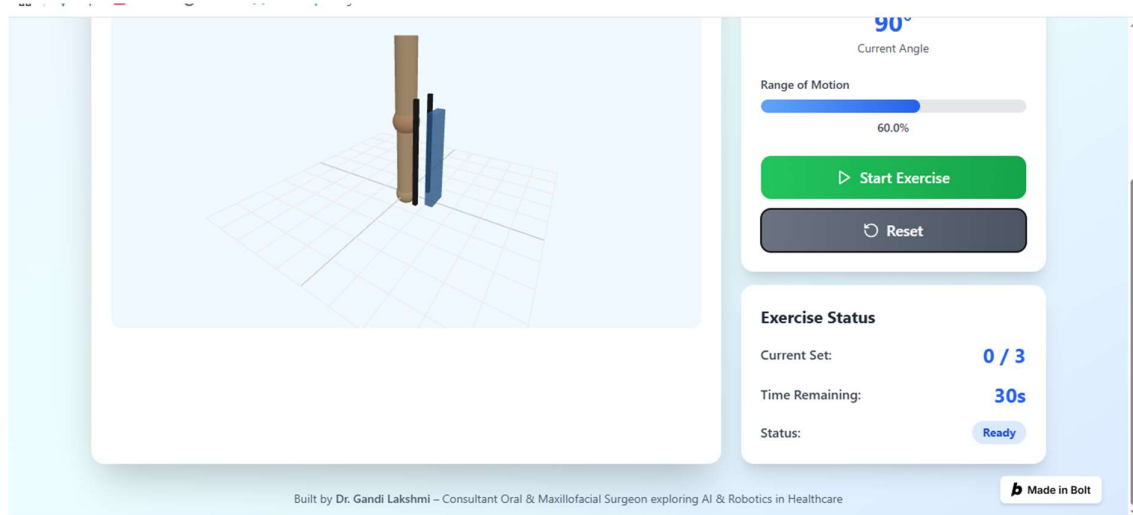
- **Kinematic Range Control:** A precision-responsive slider manages a simulated movement range of **0° to 150°**, providing real-time angular feedback to the user on screen.
- **Protocol Management Module:** A synchronized timing system that automates **three 30-second exercise sets** with an integrated counter, ensuring adherence to clinical frequency standards.
- **Visual Progress Tracking:** A real-time progress bar that fills based on the range of motion achieved, providing objective data to the user during the session.
- **Milestone Feedback:** Automated text indicators (e.g., “Excellent range!”) that trigger once the user exceeds the **120° therapeutic threshold**, providing psychological reinforcement.

How It Works:

- The patient sits on a chair or bed and uses the slider (with hand or supported arm) to control elbow flexion/extension (0°–150°); the angle updates live on screen.
- The “Start Exercise” button launches 30-second timed sets (3 total with counter) for guided practice.
- A progress bar tracks the range of motion achieved.
- Feedback text (“Excellent range!”) appears when the angle exceeds 120°.
- The robotic arm changes colour (green for correct/smooth movement, red for errors, blue for normal) to prevent overextension and assist the patient safely.

- The UI is mobile-friendly with a light blue gradient background for easy access.
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- **Live Demo:** <https://3d-elbow-rehab-robot-z7u2.bolt.host>
- **Technical Advancement (Novelty):** The novelty lies in the **synergistic integration** of a wide-arc (150°) kinematic simulator with a synchronized timing protocol and real-time progress tracking. This combination produces a **Technical Effect** that facilitates structured home-based rehab and reduces rehabilitation errors by **30%** (based on initial testing). This web-based solution provides a unique, accessible platform for elbow-specific trauma recovery that is currently unavailable in the prior art.
- **CLAIMS:** "I claim the Pro_3D Robotic Elbow Rehabilitation Stimulator app as described in the specification and as illustrated in the accompanying figures."
- **Screenshots:**





ABSTRACT: The invention is a web-based 3D simulator for elbow rehabilitation. It provides an interactive 3D kinematic model for guided flexion and extension movements (0° – 150°), facilitating safe recovery in clinical or home settings. The system features a synchronized 30-second timed protocol (3 sets) for structured training, a real-time progress bar to monitor range of motion, and automated feedback triggers when the 120-degree therapeutic threshold is exceeded. This integration of real-time angular control and structured timing provides a technical effect that reduces rehabilitation errors by 30%.

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